



Sanjeev Krishnaa Elavarasu

Roll No.: 230101089

B.Tech - Computer Science and Engineering

Indian Institute Of Technology, Guwahati

+91-7975431003

e.sanjeev@iitg.ac.in

sanjeevkrishnaa@gmail.com

linkedin.com/in/sanjeev-krishnaa

github.com/sanjeevkrishnaa

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	9.29 (Current)	2023-Present
Senior Secondary	CBSE Board	97.6%	2023
Secondary	CBSE Board	94.0%	2021

EXPERIENCE

- Summer Internship**, Elpis IT Solutions Pvt Ltd May - Jun 2024
 - Optimized Ethernet communication using the **Lightweight IP (LWIP)** protocol suite on the **Microchip SAMA5D4** embedded processor, enabling high-speed data transfer in a resource-constrained real-time environment.
 - Implemented **Direct Memory Access (DMA)** to offload data movement from the CPU, significantly enhancing throughput from **1.5 Mbps** to **20 Mbps** and reducing processing overhead in time-critical networking applications.
 - Engineered a buffer system with **scatter-gather DMA** for low-latency, high-throughput packet handling.

PROJECTS

- Algorithmic Trading Strategy for BTC and ETH Markets** Nov - Dec 2024
Gold Medal, Inter IIT Tech Meet 13.0, IIT Bombay tinyurl.com/BTC-ETH
 - Developed trading algorithms for **BTC** and **ETH** markets, leveraging their strong **correlation** and **interdependence**.
 - Designed and trained a **Q-Learning** based **Reinforcement Learning** agent to optimize trading decisions for **BTC**, achieving an annualized return of **49.97%**, with a **64.52%** win rate, and a maximum drawdown of **-13.50%**.
 - Performed change-point analysis using the **CUSUM** indicator and assessed stock data persistence with the **Hurst exponent** for **ETH**, achieving a **196.07%** annualized return, with a maximum drawdown of **-17.14%**.
- Quantitative EDA and Intraday Trading in the GB Power Market** Jan - Feb 2025
Bronze Medal, Kriti 2025, Inter Hostel Competition, IIT Guwahati tinyurl.com/GB-elec-mkt
 - Modeled volatility regimes using a **Markov Switching Model** to capture market dynamics and price fluctuations.
 - Analyzed **correlations** between spot electricity prices and renewable energy generation, weather patterns, etc.
 - Developed an intraday trading strategy based on **regime identification** and intraday price patterns, aligning trade execution with prevailing market conditions, **volatility regimes**, and **price momentum** shifts in the Energy Market.
- Market-Neutral Mean-Reversion Trading Strategy** Sep - Oct 2024
Gold Medal, Inter IIT Bootcamp, Techboard, IIT Guwahati tinyurl.com/Mean-Rev
 - Developed a **market-neutral mean-reversion** strategy for **NIFTY50** stocks, using a **momentum ranking system**.
 - Leveraged **dynamically-weighted, self-correcting** RSI & MACD scores to identify optimal investment opportunities.
 - Implemented **dynamic time-based** stop-losses, capping drawdown at **23.3%**, with a **17.27%** annualized return.

COMPETITIVE PROGRAMMING

- Codeforces**: Maximum Rating - **1787** | **Expert** | Solved **600+** problems | Handle - **RXT_SKape**
 - Codeforces Round 174 (Edu Division 2): Secured a **Global Rank of 431** among 17,000+ participants.
 - Codeforces Round 1012 (Division 2): Secured a **Global Rank of 467** among 19,000+ participants.
- Codechef**: Maximum Rating - **2058** | **5 Stars** | **Country Rank - 744** | Handle - **rxt_skape**
 - Codechef Starters 168 (Division 4): Secured a **Global Rank of 22** among 28,000+ participants.
 - Codechef Starters 170 (Division 2): Secured a **Global Rank of 73** among 2,000+ participants.
- ICPC India Preliminary Round 2024**: Secured a **Team Rank of 189** among 3,000+ teams - **Rank 2** in the batch.
- Meta Hacker Cup 2024**: Secured a **Global Rank of 1628** in Round 2, among 22,000+ overall participants.
- ICPC de Tryst 2025**: Secured a **National Rank of 83** in the ICPC style coding contest organized by **IIT Delhi**.

ACHIEVEMENTS

- Inter IIT Tech Meet 13.0**, Secured **Gold Medal** in the Quant Finance PS among teams from **22 IITs**. 2024
- Inter IIT Tech Bootcamp**, Secured **Rank 1** in the Quant Finance Problem Statement. 2024
- Kriti - Inter Hostel Tech Competition**, Secured **Rank 1** in the Optimisation Problem Statement. 2025
- Kriti - Inter Hostel Tech Competition**, Secured **Rank 3** in the Quant Finance Problem Statement. 2025
- Mimamsa 2024**, The North-East zone, **Zonal Toppers** in Mimamsa 2024, conducted by **IISER Pune**. 2024
- KCET (Karnataka Common Entrance Test)**, Secured **Rank 11** among **2,50,000+** candidates. 2023
- Joint Entrance Examination Mains**, Secured **All India Rank 1035** among **11,00,000+** candidates. 2023
- Joint Entrance Examination Advanced**, Secured **All India Rank 1283** among **2,00,000+** candidates. 2023

TECHNICAL SKILLS

- Programming**: C, C++, Python*, JavaScript*
 - Finance**: Algorithmic Trading, Portfolio Optimization
 - Web Development**: HTML, CSS, Node.js*, Express.js*
 - Operating Systems**: Windows, Linux*
 - Database Management**: MySQL
 - Miscellaneous**: Git, Verilog*, LaTeX*
- * Elementary proficiency

KEY COURSES TAKEN

- Computer Science**: Data Structures & Algorithms, Design & Analysis of Algorithms, Database Management Systems, Computer Architecture & Organization*, Digital Design, Introduction to Computing*
 - Mathematics**: Probability Theory & Random Processes*, Number Theory & Algebra, Discrete Mathematics
- * Awarded **AS/AA** Grade

POSITION OF RESPONSIBILITY

- Quant Head**, Finance and Economics Club, IIT Guwahati Mar 2025 - Present
 - Organized campus and national finance events, contributing to projects, mentorships, and workshops.